

AYMANE AIT DADS

AI Engineer Intern | Agentic AI · LLM Fine-Tuning · MLOps & Model Deployment

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and implementing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer, demonstrating full ML model development ownership. Reduced manual network analysis time by 60% at Orange Maroc by deploying a production-grade Random Forest and K-Means pipeline across 100K+ records, translating directly into network optimization team recommendations. Proficient in Python, PyTorch, TensorFlow, LangChain, and MCP server integration - bringing hands-on machine learning workflow experience and agentic AI pipeline design skills directly applicable to Ensign's AIOps security environment. Aims to contribute to the design and assessment of Agentic AI for Cyber Security, optimizing AI models for operational deployment within Ensign's threat intelligence and AIOps infrastructure.

CORE COMPETENCIES

Agentic AI Design | Python Development | Machine Learning Workflow | LangChain & LangGraph | AI Model Optimisation | MLOps & Monitoring | PyTorch / TensorFlow | MCP Servers

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and implemented a clustering pipeline using K-Means to map customer performance profiles to **5 commercial service tiers**, output adopted directly by the network optimization team.
- Built a Random Forest classification model on **100K+ fiber-optic network records** (upload speed, latency, jitter) to assess and classify network quality across thousands of nodes.
- Optimised the machine learning workflow via automated feature engineering and model evaluation, reducing **manual analysis time by ~60%** and accelerating operational decision-making.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, communicating **AI model results** to non-technical audiences in a production setting.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 · 2025 · 1st in Class Leaderboard

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head and dual learning-rate strategy across 250K training images from 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; built a 3-model ensemble with 10-view TTA (flips, crops, blur, resizes), finding that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC) for out-of-distribution generalisation.

PyTorch · OpenCLIP · ViT · AdamW · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | TensorFlow | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

Agentic AI & LLM Tooling: LangChain | LangGraph | MCP Servers | LoRA / QLoRA | RAG | Prompt Engineering | Claude API

MLOps & Data Science: Python | SQL | Git | Docker | FP16/BF16 Mixed Precision | Pandas | Power BI | PostgreSQL